MINOS-Bench: Supplementary Materials

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Contents

1 Semiconductor Domain Knowledge Base

This appendix provides the complete listing of 431 technical terms in our semiconductor domain knowledge base, organized into seven categories with associated precision weights.

1.1 Manufacturing Processes (60 terms, weight = 1.5)

Core fabrication techniques: wafer, lithography, etching, CVD, PVD, ALD, ion implantation, annealing, CMP, cleaning, inspection, metrology, mask, exposure, develop, strip, deposition, diffusion, oxidation, epitaxy, packaging, wire bonding, flip chip, WLP, SiP, BGA, CSP

Advanced lithography: immersion lithography, multi-patterning, self-aligned, spacer, hard mask, anti-reflective coating, photoresist, reticle

Etching techniques: reactive ion etching, RIE, plasma etching, dry etching, wet etching, isotropic etching, anisotropic etching

Deposition methods: PECVD, LPCVD, MOCVD, MBE, electroplating, sputtering, atomic layer deposition, physical vapor deposition

CMP and cleaning: chemical mechanical planarization, post-CMP cleaning, megasonic cleaning, RCA clean, HF dip

Metrology: overlay metrology, critical dimension measurement, ellipsometry, optical CD, scatterometry

1.2 Materials Science (51 terms, weight = 1.3)

Substrate materials: silicon, GaAs, GaN, SiC, InP, germanium, polysilicon, monocrystalline, amorphous

Dielectric materials: high-k, low-k, SiO2, Si3N4, Al2O3, HfO2, silicon dioxide, silicon nitride, hafnium oxide, aluminum oxide, zirconium oxide, tantalum pentoxide, ultra-low-k, porous silica, carbon-doped oxide

Metal materials: copper, aluminum, tungsten, titanium, tantalum, indium, silver, gold, platinum, nickel, cobalt, ruthenium, molybdenum

Compound semiconductors: gallium arsenide, gallium nitride, silicon carbide, indium phosphide, aluminum nitride, zinc oxide

2D materials: graphene, MoS2, transition metal dichalcogenides

Gate stack materials: metal gate, work function metal, barrier layer, capping layer, etch stop layer

1.3 Device Physics (62 terms, weight = 1.4)

Transistor types: MOSFET, FinFET, GAAFET, CMOS, BJT, JFET, IGBT, diode, transistor

Advanced transistors: gate-all-around, nanosheet, nanowire, CFET, complementary FET, vertical transistor, TFET, tunnel FET, FDSOI, fully depleted SOI

Memory devices: memory, DRAM, SRAM, NAND, NOR, FRAM, ReRAM, MRAM, flash memory, 3D NAND, PCM, phase change memory, RRAM, STT-MRAM, emerging memory

Passive components: resistor, capacitor, inductor

Integrated circuits: IC, integrated circuit, ASIC, SoC, system on chip, logic circuit, analog circuit, mixed-signal

Sensors and MEMS: MEMS, sensor, actuator, image sensor, CMOS sensor, pressure sensor, accelerometer, gyroscope

Specialty devices: RF, power device, optoelectronic, LED, laser diode, photodiode, solar cell, TFT, thin film transistor

1.4 Process Nodes (65 terms, weight = 1.6)

Process nodes: 14nm, 10nm, 7nm, 5nm, 3nm, 2nm, 1nm, 28nm, 22nm, 16nm, 12nm, 8nm, 6nm, 4nm, process node, technology node

Lithography technologies: EUV, DUV, 193nm, 248nm, 365nm, ArF, KrF, i-line, g-line, extreme ultraviolet, deep ultraviolet, 13.5nm, high-NA EUV

Process integration: FinFET process, SOI, FD-SOI, bulk, bulk silicon, strained silicon, strain engineering, SiGe, silicon germanium, HEMT, high electron mobility

Process stages: FEOL, BEOL, MEOL, front end of line, back end of line, middle of line

Advanced techniques: gate-first, gate-last, replacement gate, high-k metal gate, HKMG, multiple patterning, SADP, SAQP, self-aligned double patterning, self-aligned quadruple patterning, LELE, LFLE

Interconnect: damascene, dual damascene, copper interconnect, via, through silicon via, TSV, backside power delivery

1.5 Testing Methodologies (51 terms, weight = 1.2)

Reliability testing: reliability, yield, failure analysis, electrical test, functional test, burn-in, temperature cycling, thermal shock, humidity test

Failure mechanisms: electromigration, hot carrier, NBTI, TDDB, negative bias temperature instability, time dependent dielectric breakdown, stress migration, void formation

Quality control: SPC, statistical process control, parameter drift, defect density, process capability, Cpk, design of experiments, DOE

Inspection techniques: critical dimension, CD-SEM, AFM, SEM, TEM, X-ray, atomic force microscopy, scanning electron microscopy, transmission electron microscopy, optical inspection, e-beam inspection

Electrical characterization: I-V curve, C-V measurement, DLTS, deep level transient spectroscopy, Hall effect, mobility measurement, sheet resistance, four-point probe, contact resistance

Wafer-level testing: probe test, wafer sort, parametric test, functional test, final test, package test

1.6 Applications (62 terms, weight = 1.1)

Computing processors: AI chip, GPU, CPU, NPU, TPU, FPGA, ASIC, graphics processing unit, neural processing unit, tensor processing unit, application specific IC, DSP, digital signal processor, MCU, microcontroller

Computing systems: edge computing, cloud computing, data center, server, supercomputer, quantum computing, neuromorphic computing, in-memory computing

Consumer electronics: smartphone, tablet, laptop, wearable, smartwatch, AR glasses, VR headset

Automotive: automotive electronics, ADAS, autonomous driving, LiDAR, radar, infotainment, powertrain

Communication: 5G, 6G, IoT, internet of things, wireless, RF transceiver, baseband, modem

AI/ML: neural network, machine learning, deep learning, inference, training, transformer, large language model

Emerging applications: blockchain, cryptocurrency, mining, virtual reality, VR, augmented reality, AR, metaverse, HPC, high performance computing

1.7 Performance Parameters (80 terms, weight = 1.7)

Voltage parameters: threshold voltage, Vth, supply voltage, VDD, VSS, voltage, operating voltage, breakdown voltage, junction temperature

Current parameters: leakage current, drive current, Ion, Ioff, current, saturation current, subthreshold current, gate leakage, junction leakage

Power parameters: power consumption, static power, dynamic power, power dissipation, TDP, thermal design power, power density, energy efficiency

Timing parameters: switching speed, delay, propagation delay, rise time, fall time, setup time, hold time, clock skew, clock frequency, access time

Frequency parameters: frequency, bandwidth, cutoff frequency, fT, fmax, operating frequency, clock speed

Resistance and capacitance: resistance, capacitance, Ron, on-resistance, gate capacitance, parasitic capacitance, RC delay, interconnect resistance

Temperature parameters: operating temperature, junction temperature, thermal resistance, temperature coefficient

Noise and linearity: noise, noise figure, SNR, signal to noise ratio, linearity, THD, total harmonic distortion

RF parameters: gain, phase, S-parameters, impedance matching, insertion loss, return loss

Performance metrics: DIBL, drain induced barrier lowering, subthreshold swing, SS, transconductance, gm, output conductance, GIDL, gate induced drain leakage, mobility, carrier mobility, saturation velocity

1.8 Knowledge Base Design Rationale

Coverage breadth: The 431 terms span the full semiconductor technology stack from materials science to system applications.

Precision weighting: Category weights ($w_i \in [1.1, 1.7]$) reflect technical rigor requirements, with parameters (1.7) and processes (1.6) weighted highest due to quantitative precision requirements.

Term granularity: Includes both high-level concepts (e.g., "transistor") and specific implementations (e.g., "FinFET", "GAAFET", "nanosheet").

Temporal coverage: Terms encompass mature technologies, current leading-edge, and emerging concepts to ensure benchmark relevance as technology advances.

2 Generation System Implementation

2.1 Precision Classification

Documents are classified based on technical density $\rho(d)$ and weighted high-precision category presence:

$$p(d) = \begin{cases} \text{high} & \text{if } \rho(d) > 0.20 \text{ or } \omega_h(d) > 8 \\ \text{medium} & \text{if } \rho(d) > 0.12 \text{ or } \omega_h(d) > 4 \\ \text{low} & \text{otherwise} \end{cases} \tag{1}$$

where $\omega_h(d) = \sum_{c_i \in \{\text{parameters,processes}\}} w_i \cdot |\{t \in T_{c_i} : n(t,d) > 0\}|$. The disjunctive logic ensures documents containing critical parameters are classified as high-precision even with low overall density.

2.2 Adaptive Temperature Control

Base Temperature Ranges. Temperature ranges vary by precision level to balance accuracy with diversity:

$$[\tau_{\min}(p), \tau_{\max}(p)] = \begin{cases} [0.4, 0.8] & p = \text{high} \\ [0.5, 0.9] & p = \text{medium} \\ [0.6, 1.0] & p = \text{low} \end{cases} \tag{2}$$

Progressive Temperature. Base temperature progresses with successful generations to encourage diversity:

$$\tau_{\text{progress}}(k) = (\tau_{\max} - \tau_{\min}) \times \min\left(\frac{k}{20}, 0.8\right) \tag{3}$$

where k is the count of successfully generated questions.

Category-Specific Adjustments. Fine-grained adjustments based on dominant content category:

$$\Delta\tau_c(c^*) = \begin{cases} -0.10 & c^* = \text{parameters} \\ -0.08 & c^* = \text{processes} \\ -0.05 & c^* = \text{devices} \\ -0.03 & c^* = \text{testing} \\ 0.00 & c^* = \text{materials} \\ +0.02 & c^* = \text{manufacturing} \\ +0.05 & c^* = \text{applications} \end{cases} \tag{4}$$

Failure Recovery Boosts. Temperature increases after generation failures:

$$\Delta\tau_{\text{attempt}}(a) = \min(0.25, 0.08 \times a) \tag{5}$$
$$\Delta\tau_{\text{similarity}}(s) = \min(0.15, 0.05 \times s) \tag{6}$$

where a counts all failed attempts and s tracks consecutive similarity failures.

Final Temperature Computation.

$$\tau = \text{clip}\left(\tau_{\min} + \tau_{\text{progress}} + \Delta\tau_c + \Delta\tau_{\text{attempt}} + \Delta\tau_{\text{similarity}}, 0.1, 1.0\right) \tag{7}$$

Complementary Nucleus Sampling. Adaptive top-p maintains coherence during high-temperature exploration:

$$p_{\text{nucleus}}(\tau) = \begin{cases} 0.95 & \tau \leq 0.4 \\ 0.90 & 0.4 < \tau \leq 0.7 \\ 0.85 & \tau > 0.7 \end{cases} \tag{8}$$

2.3 Adaptive Similarity Thresholds

Base Thresholds by Precision Level.

$$\theta_{\text{base}}(p) = \begin{cases} 0.70 & p = \text{high} \\ 0.75 & p = \text{medium} \\ 0.80 & p = \text{low} \end{cases} \tag{9}$$

Lower thresholds for high-precision content acknowledge that questions about specialized topics necessarily share technical terminology while remaining substantively different.

Progressive Relaxation.

$$\theta_{\text{sim}}(r) = \max(0.50, \theta_{\text{base}}(p) - 0.05 \times r) \tag{10}$$

Weighted Similarity Computation.

$$\text{sim}_{\text{weighted}}(q, q') = \text{Jaccard}(W_q, W_{q'}) + \alpha \cdot \text{Jaccard}(T_q, T_{q'}) \tag{11}$$

where W denotes all word tokens, $T \subseteq W$ denotes technical terms, and $\alpha = 0.05$.

For corpora with ≥ 5 existing questions, TF-IDF semantic similarity provides additional validation:

$$\text{sim}_{\text{semantic}}(q, q') = \cos(\vec{v}_{\text{TF-IDF}}(q), \vec{v}_{\text{TF-IDF}}(q')) \tag{12}$$

Questions are rejected if $\max_{q' \in \mathcal{Q}} \text{sim}_{\text{semantic}}(q, q') > \theta_{\text{sim}} + 0.05$.

2.4 Document Chunking

Documents are segmented using sliding window approach: chunk size 512 tokens, overlap 128 tokens (25%). Cross-document linking threshold: $\theta_{\text{link}} = 0.1$.

3 Dynamic Prompt Templates

This section describes the core logic of our dynamic prompt engineering system.

3.1 Prompt Architecture

All prompts consist of four modular components that combine based on generation context:

1. **Base Template**: Defines role (AI Benchmark Scientist), injects context sections, specifies JSON output format
2. **Test-Type Module**: Scenario-specific instructions based on evaluation target
3. **Format Specification**: Question type requirements with balance tracking
4. **Diversity Directive**: Activated after repeated similarity failures

3.2 Test-Type Configurations

Three test types target different RAG capabilities:

Table 1: Test-type configurations

Test Type	Instruction Focus	Primary	Secondary
Robustness	Find rare/specific facts that are easily overlooked (needle-in-haystack); construct unambiguous ground truth	Factuality	Depth, Completeness
Multi-Hop	Identify logical chains requiring ≥ 2 sections; force cross-section synthesis	Completeness	Depth, Factuality
Generation	Assume perfect context; require deep reasoning (analysis, comparison, causation)	Depth	Completeness, Factuality

3.3 Question Format Balance

We maintain target distribution of 40% objective and 60% subjective questions. The system tracks:

$$r_{\text{obj}} = \frac{n_{\text{obj}}}{n_{\text{obj}} + n_{\text{subj}}} \tag{13}$$

When $r_{\text{obj}} < 0.5$, the system forces objective question generation.

Objective Questions (40% target). Four subtypes with specific requirements:

- **Mathematical Calculation (30%)**: Must extract ALL parameters from context; use context-specific formulas; show step-by-step with units
- **Fill-in-Blank (25%)**: Extract EXACT values with conditions and units as stated
- **True/False (25%)**: Statements about specific mechanisms requiring multi-step understanding; create both true and false statements
- **Multiple Choice (20%)**: All options in one line; only context provides distinguishing details

Subjective Questions (60% target). Five archetypes rotate via (question_count) $\mod 5$:

1. Definition/Specification: specific definitions or specification values
2. Process Explanation: process sequences or event chains
3. Causal Reasoning: reasons or purposes behind specifications
4. Comparative Analysis: compare/contrast related concepts
5. Problem Identification: potential problems, defects, or limitations

3.4 Diversity Enhancement

When similarity failures exceed threshold ($s > 2$), a lightweight directive is injected:

> *DIVERSITY FOCUS: Previous s uniqueness attempts suggest trying a new approach*
> *RECOMMENDED STYLE: Focus on "[current archetype from rotation]"*

This provides guidance to explore unexplored patterns without over-constraining generation.

3.5 Expert Validation of Generated Benchmarks

To assess generation quality, three domain experts rated 120 QA pairs (30 per condition) from a 2×2 ablation study (adaptive temperature $\times$ adaptive similarity threshold) on four dimensions (1–5 scale). As shown in Table 2, the full adaptive system (Group D) tends to produce higher-quality outputs across all evaluated dimensions compared to single-component variants, suggesting that the two mechanisms are complementary.

Table 2: Expert evaluation of QA generation quality across ablation conditions. Each group generated 30 QA pairs (120 total), rated by three domain experts on a 1–5 scale.

Group	**Configuration**	**Acc.**	**Rel.**	**Diff.**	**Div.**	**Retries**	**Time**
A (Baseline)	Fixed τ, Fixed θ	4.67	4.72	4.00	3.67	15	18m 33s
B (+Adaptive τ)	Adaptive τ, Fixed θ	4.89	4.83	4.17	3.39	16	30m 37s
C (+Adaptive θ)	Fixed τ, Adaptive θ	4.61	4.61	3.61	3.28	47	36m 55s
D (Full System)	Adaptive τ, Adaptive θ	**5.00**	**5.00**	**4.60**	**4.53**	**0**	25m 15s

4 Evaluation Rubrics

All metrics use GPT-4.1-mini via DeepEval's G-Eval with chain-of-thought reasoning, scoring 0–10 (internally normalized to 0–1). Rubrics adapt between objective and subjective questions.

4.1 Completeness

Table 3: Completeness rubric

Score	Objective	Subjective
9–10	Correct answer with complete explanation and clear reasoning	Addresses all key aspects thoroughly
7–8	Correct answer with good explanation, minor gaps	Covers all major aspects with good detail
5–6	Answer with basic explanation, some missing details	Addresses most important aspects reasonably
3–4	Partial answer or incomplete explanation	Covers key aspects with varying depth
1–2	Minimal answer, lacks proper explanation	Addresses some but misses important points
0	No clear answer or off-topic	Fails to address key aspects

4.2 Depth

Table 4: Depth rubric

Score	Objective	Subjective
9–10	Sophisticated understanding with detailed methodology	Sophisticated understanding with nuanced analysis
7–8	Strong analytical thinking with clear solution steps	Strong analytical thinking with good conceptual depth
5–6	Reasonable technical detail with some analysis	Reasonable analysis with adequate technical detail
3–4	Basic approach but relatively shallow analysis	Surface-level analysis with limited sophistication
1–2	Minimal analytical content, superficial approach	Superficial treatment with minimal content
0	No meaningful analysis demonstrated	No meaningful analysis or understanding

4.3 Factuality

Table 5: Factuality rubric

Score	With Retrieval Context	Without Context
9–10	All claims match context exactly, no contradictions	Claims match expected output, no contradictions
7–8	Minor imprecision in non-critical details	Most claims consistent, no significant errors
5–6	Generally accurate with minor factual gaps	Some inconsistencies but generally reasonable
3–4	Noticeable inaccuracies but core info correct	Notable contradictions or implausible info
1–2	Several factual errors or contradictions	Major contradictions or obviously incorrect
0	Major errors or predominantly false information	—

Table 6: Relevance rubric

Score	Objective	Subjective
9–10	Directly addresses specific question format	Every sentence directly addresses the question
7–8	Stays focused with minimal deviation	Strongly focused with minimal deviation
5–6	Generally addresses question, some unnecessary details	Mostly focused with minor tangential elements
3–4	Partially addresses with irrelevant information	Generally on-topic but includes unnecessary info
1–2	Somewhat related with significant drift	Partially addresses with significant drift
0	Fails to address specific requirements	Minimal connection to actual question

4.4 Relevance

4.5 Context Utilization

Table 7: Context utilization rubric

Score	With Retrieval Context	Manual Setup
9–10	Seamlessly weaves multiple sources together	Sophisticated domain expertise with advanced terminology
7–8	Effectively uses most relevant context	Clear evidence of domain-specific knowledge
5–6	Incorporates key context adequately	Some evidence of specialized knowledge
3–4	Uses some context, integration could improve	Minimal evidence of specialized knowledge
1–2	Minimal effective use of available context	Generic response with limited domain content
0	Completely ignores or contradicts context	No domain-specific content

4.6 Support Quality

Table 8: Support quality rubric

Score	Objective	Subjective
9–10	Exceptional evidence with specific calculations/formulas	Exceptional evidence with specific, concrete details
7–8	Strong evidence with good specificity	Strong evidence with good specificity
5–6	Adequate details with reasonable explanation	Adequate details with reasonable examples
3–4	Some details but could be more specific	Some details but could be more specific
1–2	Minimal evidence or poorly explained	Minimal evidence or poorly chosen examples
0	Lacks evidence or contains misleading explanations	Lacks evidence or contains misleading examples

5 Evaluation Platform

5.1 Framework Capabilities

Table 9: RAG Framework Capabilities

Framework	API Upload	Sources Exposed	Model Config	Setup
AnythingLLM	✓	✓	✓	Recommended
RAGFlow	✓	✓	✓	Recommended
MaxKB	–	–	✓	Required
Metaso	✓	Partial	–	Required

Key Differences. **AnythingLLM** provides full retrieval transparency through its sources field, returning document chunks with relevance scores.

RAGFlow uses an OpenAI-compatible API endpoint, returning responses in standard format. Source attribution requires inference from response content.

MaxKB does not expose retrieval context through its API—sources are visible only in the web interface. Our adapter implements heuristic detection using domain-specific indicators (e.g., "according to", "standard specifies", "SEMI") and professional terminology presence.

Metaso returns structured references but uses a proprietary model that cannot be reconfigured, limiting participation to cross-framework evaluation.

5.2 Adaptive Evaluation Strategy

The evaluation layer adapts metric computation based on framework capabilities. For frameworks without source exposure, Context Utilization switches from direct source comparison to heuristic inference based on response characteristics, ensuring fair evaluation across frameworks with different transparency levels.

5.3 Dual-Mode Assessment

- **Mode A (with_kb)**: Standard RAG query through framework's native retrieval pipeline
- **Mode B (without_kb)**: Gold context injected into prompt, bypassing retrieval to isolate generation capabilities

Cross-mode score differences (Δ = Mode B − Mode A) enable failure attribution: large positive Δ indicates retrieval failures; consistently low scores in both modes indicate generation-stage weaknesses.

6 Complete Experimental Results

Table 10: DeepSeek-v3.2-Exp performance across context configurations.

Config	Fact.	Depth	Comp.	Rel.	Ctx.U.	Supp.	Avg.
(4K, 1K)	0.626	0.554	0.592	0.627	0.783	0.534	0.619
(8K, 2K)	0.632	0.602	0.640	0.661	0.804	0.594	0.656
(10K, 4K)	0.663	0.606	0.651	0.669	0.814	0.609	0.669
(12K, 4K)	0.690	0.640	0.688	0.701	0.824	0.644	0.698
(14K, 4K)	0.744	0.693	0.752	0.740	0.845	0.709	0.747
(16K, 4K)	0.792	0.718	0.783	0.781	0.868	0.735	0.780
(18K, 4K)	0.837	0.755	0.820	0.813	0.884	0.776	0.814
(20K, 4K)	0.865	0.796	0.860	0.850	0.901	0.823	0.849
(24K, 4K)	0.868	0.812	0.880	0.857	0.915	0.839	0.862
(28K, 4K)	0.892	0.811	0.879	0.868	0.916	0.842	0.868
(32K, 4K)	**0.904**	**0.838**	**0.898**	**0.875**	0.912	**0.872**	**0.883**

Table 11: Qwen-Plus performance across context configurations.

Config	Fact.	Depth	Comp.	Rel.	Ctx.U.	Supp.	Avg.
(4K, 1K)	0.663	0.662	0.686	0.677	0.805	0.641	0.689
(8K, 2K)	0.665	0.710	0.724	0.704	0.801	0.696	0.717
(10K, 4K)	0.680	0.736	0.764	0.736	0.828	0.724	0.745
(12K, 4K)	0.697	0.749	0.765	0.722	0.823	0.747	0.751
(14K, 4K)	0.723	0.772	0.804	0.760	0.834	0.767	0.777
(16K, 4K)	0.750	0.789	0.809	0.768	0.848	0.770	0.789
(18K, 4K)	0.732	0.783	0.813	0.774	0.844	0.783	0.788
(20K, 4K)	0.778	0.810	0.830	0.798	0.852	0.801	0.812
(24K, 4K)	0.833	0.829	0.861	0.827	0.882	0.835	0.845
(28K, 4K)	**0.866**	**0.849**	**0.894**	**0.859**	**0.916**	**0.857**	**0.874**
(32K, 4K)	0.855	0.829	0.867	0.847	0.889	0.831	0.853

Table 12: Gemini-2.5-Flash performance across context configurations.

Config	Fact.	Depth	Comp.	Rel.	Ctx.U.	Supp.	Avg.
(4K, 1K)	0.600	0.374	0.399	0.449	0.669	0.350	0.474
(8K, 2K)	0.643	0.536	0.563	0.583	0.757	0.523	0.601
(10K, 2K)	0.666	0.550	0.595	0.622	0.758	0.622	0.636
(12K, 4K)	0.701	0.641	0.692	0.708	0.819	0.651	0.702
(14K, 4K)	0.743	0.655	0.709	0.726	0.839	0.656	0.721
(16K, 4K)	0.760	0.669	0.733	0.731	0.816	0.676	0.731
(18K, 4K)	0.776	0.710	0.771	0.784	0.847	0.728	0.769
(20K, 4K)	0.833	0.738	0.804	0.808	0.878	0.752	0.802
(24K, 4K)	0.854	0.765	0.827	0.837	0.882	0.790	0.826
(28K, 4K)	**0.901**	**0.818**	**0.886**	**0.890**	**0.907**	**0.852**	**0.876**
(32K, 4K)	0.869	0.775	0.844	0.836	0.889	0.801	0.836

Table 13: Qwen-2.5-72B-Instruct with extended context windows.

Config	Fact.	Depth	Comp.	Rel.	Ctx.U.	Supp.	Avg.
(4K, 1K)	0.511	0.568	0.603	0.589	0.758	0.536	0.594
(8K, 2K)	0.544	0.590	0.629	0.605	0.785	0.560	0.619
(16K, 4K)	0.643	0.661	0.713	0.692	0.798	0.630	0.690
(32K, 4K)	0.794	0.762	0.827	0.812	0.852	0.764	0.802
(64K, 6K)	0.801	0.759	0.820	0.796	0.856	0.738	0.795
(128K, 8K)	0.797	0.766	0.829	0.812	**0.863**	0.762	**0.805**

Table 14: Cross-model comparison at key configurations.

Model	4K	16K	28K	32K
DeepSeek-v3.2-Exp	0.619	0.780	0.868	**0.883**
Qwen-Plus	**0.689**	**0.789**	**0.874**	0.853
Gemini-2.5-Flash	0.474	0.731	0.876	0.836
Qwen-2.5-72B	0.594	0.690	—	0.802

Key Findings. Three scaling behaviors emerge:

- **Logarithmic Growth (DeepSeek):** Consistent improvement from 4K (0.619) to 32K (0.883) with no saturation or decline.
- **Early Strength, Late Saturation (Qwen-Plus):** Highest at 4K (0.689), peaks at 28K (0.874), declines at 32K (0.853).
- **Cold-Start Dynamics (Gemini):** Lowest at 4K (0.474), peaks at 28K (0.876), declines at 32K (0.836).

The 28K→32K decline observed in Qwen-Plus and Gemini suggests information overload at extreme context lengths, while DeepSeek's continued improvement indicates superior noise tolerance mechanisms.